Prodigy Anesthesia Quick Review

Tobacco Abuse

What is the concept of pack-years?	Pack-years is a unit of measurement used to quantify how much a person has smoked during their lifetime. For example, a patient who smokes a pack a day for 10 years has smoked 10 pack-years. Because a single pack contains 20 cigarettes, a patient who smokes 20 cigarettes a day for 50 years has smoked 50 pack-years. Likewise, a patient who smokes twice as much in half the time (40 cigarettes in 25 years) has also smoked 50 pack-years. *Khan BJ, et al.. Smoking and age related macular degeneration: the number of pack years of cigarette smoking is a major determinant of risk for both geographic atrophy and choroidal neovascularisation. [Online] June 24, 2008. <http://bjo.bmj.com/cgi/content/abstract/90/1/75>.*
Does chronic smoking produce a restrictive or obstructive lung disease?	Obstructive *Morgan GE, Mikhail MS, Murray MJ. Clinical Anesthesiology. 4th ed. New York, NY: McGraw-Hill; 2006: 558, 576-577.*
How does smoking affect the oxyhemoglobin curve?	Smoking shifts the oxyhemoglobin dissociation curve to the left, making oxygen less available to the tissues. *Morgan GE, Mikhail MS, Murray MJ. Clinical Anesthesiology. 4th ed. New York, NY: McGraw-Hill; 2006: 558, 576-577.*
Are there any other causes of emphysema besides smoking?	Emphysema, however, is almost always associated with smoking. In rare circumstances, young individuals may develop emphysema due to a homozygous deficiency of alpha-1 antitrypsin. *Stoelting RK, Dierdorf SF. Anesthesia & Co-Existing Diseases. 5th ed. New York, NY: Churchill-Livingston; 2008: 171-172.*
What is the minimum amount of time a patient should stop smoking prior to surgery to significantly improve mucociliary function?	Mucociliary function improves significantly at about 6 to 8 weeks of smoking cessation. Airway management may be made even more difficult prior to that amount of time as there is an increase in pulmonary secretions during the first month of smoking cessation. *Stoelting RK, Dierdorf SF. Anesthesia & Co-Existing Diseases. 5th ed. New York, NY: Churchill-Livingston; 2008: 171-172.*
What is the role of pulmonary function tests in the preoperative evaluation of smokers?	In patients who smoke, there is no evidence that pulmonary function tests are of any use in risk assessment or modification of the anesthetic treatment plan. *Ezekiel MR. Handbook of Anesthesiology 2004-2005 Ed. Laguna Hills, CA: Current Clinical Strategies; 2004: 19, 34.*

Copyright 2011 Prodigy Anesthesia. Unlawful to duplicate, distribute, or sell this document.

Prodigy Anesthesia Quick Review

Tobacco Abuse

What preoperative medications would be useful to improve pulmonary performance in smokers?	Albuterol and ipratropium bromide (Atrovent) are commonly used in COPD patients to bronchodilate and facilitate the movement of gases through the pulmonary tree. Glycopyrrolate (Robinul) is also administered in this population as an antisialogogue and tracheal drying agent. *Stoelting RK, Dierdorf SF. Anesthesia & Co-Existing Diseases. 5th ed. New York, NY: Churchill-Livingston; 2008: 171-172.*
How does smoking affect the pulmonary cilia?	The number and function of cilia decrease in smokers. Smoking causes mucociliary dysfunction including, ciliary destruction, hypokinesis, and dyskinesis. *Stoelting RK, Dierdorf SF. Anesthesia & Co-Existing Diseases. 5th ed. New York, NY: Churchill-Livingston; 2008: 171-172.*
How does the affinity of carbon monoxide to hemoglobin compare to that of oxygen?	The affinity of carbon monoxide to hemoglobin is 200-300 times greater than that of oxygen. *Morgan GE, Mikhail MS, Murray MJ. Clinical Anesthesiology. 4th ed. New York, NY: McGraw-Hill; 2006: 1044.*
What the benefits of smoking cessation prior to surgery?	Cessation for 8 weeks prior to general anesthesia produces a substantial decrease in the risk for postoperative pulmonary complications. Because normalization of mucociliary function takes 2-3 weeks and sputum production increases during that time, cessation for less than 4 weeks may increase the risk for postoperative pulmonary complications. Carbon monoxide levels can be up to 10 times normal in smokers, but can return to normal within 24 hours, thus improving oxygenation. *Barash PG, Cullen BF, Stoelting RK. Clinical Anesthesia. 6th ed. Philadelphia, PA: Lippincott Williams and Wilkins; 2009: 252.*
How does the increased plasma carboxyhemoglobin concentrations associated with recent cigarette smoking affect the accuracy of the oxygen saturation monitor?	Increased carboxyhemoglobin concentrations will cause the pulse oximeter to overestimate the SpO2 and the result will be falsely high. *Morgan GE, Mikhail MS, Murray MJ. Clinical Anesthesiology. 4th ed. New York, NY: McGraw-Hill; 2006: 141.*
How long should a patient refrain from smoking before the ability of the lungs to respond to pulmonary infection returns to normal?	Smoking interferes with normal immune response to pulmonary infection and can predispose the patient to the development of pneumonia following anesthesia. It takes about 6 weeks of smoking cessation for the immune responses of the lungs to return to normal. *Stoelting RK, Dierdorf SF. Anesthesia & Co-Existing Diseases. 5th ed. New York, NY: Churchill-Livingston; 2008: 172.*
In order to minimize the effects of carbon monoxide on oxygen-carrying capacity, a patient should be instructed to refrain from smoking for at least how many hours?	12-18 hours. The elimination half-life of carbon monoxide is about 4 to 6 hours which means that a smoke free interval of 12 to 18 hours results in marked decreases in carboxyhemoglobin levels and a corresponding increase in oxygen carrying capacity. *Stoelting RK, Dierdorf SF. Anesthesia & Co-Existing Diseases. 5th ed. New York, NY: Churchill-Livingston; 2008: 171.*

Copyright 2011 Prodigy Anesthesia. Unlawful to duplicate, distribute, or sell this document.